

15APMO1

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Problem

Let ABC be a triangle, and let D be a point on side BC . A line through D intersects side AB at X and ray AC at Y . The circumcircle of triangle BXD intersects the circumcircle ω of triangle ABC again at point Z distinct from point B . The lines ZD and ZY intersect ω again at V and W respectively. Prove that $AB = VW$.

Solution. It is enough to show that $AV \parallel BZ$. Since $DX \parallel AV$ due to Reim's, it is enough to show that $XD \parallel BZ$.

$\angle XDZ = \angle CDY$ and $\angle CBW = \angle CZW$. It is enough to show that $CDZY$ is cyclic.

$\angle ZDY = \angle ZBX$ and $\angle ZCY = \angle ZBA$. Since X lies on AB , $CDZY$ is cyclic.